

Pregunta 19

Link Github repository

<https://github.com/A01178406/Examen-final-19.git>

Link Colab

<https://colab.research.google.com/drive/10uKTDa2D-6pTmgBkXnjgN2VzDNIqVrgh?usp=sharing>

Link copilot/Grok

https://grok.com/share/c2hhcmQtMg%3D%3D_072f1e48-d47c-48cf-b6bc-eada792f8c2d

Interpretations

The Python code for question 22.25 from Basic Econometrics by Damodar N. Gujarati digs into 3-month (TB3M) and 6-month (TB6M) Treasury bill rates from January 1982 to March 2008, tackling parts (a) to (e) and throwing in a 12-month forecast as requested. It starts by grabbing data from FRED or cooking up fake data to get 315 monthly points, then plots TB3M and TB6M, showing they track closely, with TB6M a bit higher due to the term premium, hinting at a long-term link even if they wiggle around. Unit root tests (ADF) usually show both series are $I(1)$ —jumpy in levels but steady when differenced—meaning they're non-stationary but can be tamed. The Engle-Granger cointegration test likely confirms they're tied together long-term (p-value < 0.05), so they move in sync thanks to stuff like Fed policies; if they're not cointegrated, it points to market quirks or big shifts, making things trickier to predict. Cointegration means the yield curve's slope is pretty reliable, but no cointegration suggests the rates could drift apart. For the VAR model with four lags, the code uses levels if they're cointegrated to keep that long-term vibe, or differences if not, to keep things legit and avoid wonky results. It also forecasts a year ahead (April 2008–March 2009), plotting past and future values to show where rates might head, usually with TB6M above TB3M. The code fixes a pesky `SyntaxError` by making sure multi-line strings, especially in the unit root section, are closed properly, and it churns out clear plots, test results, VAR details, and forecasts that shed light on how these interest rates behave and what to expect next.